🔑 **Answer Key — Teacher Copy · Lesson 3-4**

MSTAR PRACTICE WORKSHEET

Lesson 3-4: Determine Equivalent Ratios Using Graphs

Grade 6 Mathematics · Standard 6.AT.3a

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: B

A ratio table shows (2, 6), (4, 12), and (6, 18). Which ordered pair comes next if the pattern continues?

A. (7, 20)

B. (8, 24)

C. (8, 22)

D. (10, 24)

The pattern adds 2 to x and 6 to y each time. After (6, 18): x = 6+2 = 8, y = 18+6 = 24. The next point is (8, 24).

Question 2 SELECTED RESPONSE — CORRECT: B

When you graph equivalent ratios, the points will always form what shape?

A. A curved line

B. A straight line through the origin

C. A zigzag

D. A circle

Equivalent ratios are proportional, so their graph is always a straight line that passes through the origin (0, 0).

Question 3 SELECTED RESPONSE — CORRECT: C

A ratio table shows (2, 8), (3, 12), and (5, 20). What is the y-value when x = 7?

A 21

B 24

C 28

D 35

The ratio is y/x = 8/2 = 4. For every x, y = 4x. When x = 7: y = 4 × 7 = 28.

Question 4 SELECTED RESPONSE — CORRECT: B

Which point would NOT be on the graph of the ratio 1:5?

A (2, 10)

B (3, 10)

C (4, 20)

D (5, 25)

For the ratio 1:5, every y-value should be 5 times the x-value. (3, 10): 5 × 3 = 15, not 10. So (3, 10) does not belong.

Question 5 SELECTED RESPONSE — CORRECT: D

Points from a ratio table are plotted at (1, 4), (2, 8), and (3, 12). What is the ratio of y to x for each point?

A 1:4

B 2:8

C 3:4

D 4:1

For each point: 4÷1 = 4, 8÷2 = 4, 12÷3 = 4. The ratio of y to x is always 4:1.

Question 6 SELECTED RESPONSE — CORRECT: D

Two students graph ratio tables. Student A plots (1,2), (2,4), (3,6). Student B plots (1,3), (2,6), (3,9). Whose line is steeper?

A Student A

B Same steepness

C Cannot tell

D Student B

Student A's ratio is 1:2 (y goes up 2 for each 1 in x). Student B's ratio is 1:3 (y goes up 3 for each 1 in x). A higher rate of change means a steeper line, so Student B's is steeper.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: A

A ratio table pairs 1 hour with 45 miles, 2 hours with 90 miles, and 3 hours with 135 miles. Which ordered pair belongs on the graph of this relationship?

- (A) (4, 180)
- (B) (4, 45)
- (C) (4, 140)
- (D) (180, 4)

The relationship is 45 miles per hour, so 4 hours pairs with 4 × 45 = 180 miles, giving the point (4, 180).

- **B:** 45 miles is the distance for ONE hour. At 4 hours it is 4 x 45.
- **C:** 140 is not a multiple of 45. Four hours at 45 miles per hour gives 4 x 45.
- **D:** The coordinates are swapped. Hours is the first coordinate and miles the second.

Part B — correct answer: A

What does it mean that the graph of this relationship passes through the origin (0, 0)?

- (A) When zero hours have passed, zero miles have been traveled.
- (B) The graph must be a curve.
- (C) The unit rate is zero.
- (D) The relationship only works for whole numbers.

A proportional relationship passes through the origin because zero of one quantity corresponds to zero of the other — no distance is covered in no time.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that are true about graphing equivalent ratios:

- (A) The points of a proportional relationship lie on a straight line.
- (B) The line passes through the origin (0, 0).
- (C) Each ordered pair comes from a row of the ratio table.
- (D) The points form a curve that bends upward.
- (E) The unit rate can be read as the y-value when x = 1.

A, B, C, and E are true. D is false — equivalent ratios graph as a straight line, not a curve.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A student plots the points (1, 3), (2, 6), (3, 9), and (4, 15) from a ratio table. She says they form a straight line because they go up. Is she correct? Explain how to check.

Model answer: She is not correct. While the first three points follow the pattern y = 3x (3, 6, 9), the fourth point should be (4, 12), not (4, 15). If you plot all four points, (4, 15) would not land on the line through the others. The ratio 4:15 does not simplify to 1:3 like the others.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.